Aerospace Science and Technology 144 (2024) 108804

Contents lists available at ScienceDirect

Aerospace Science and Technology

journal homepage: www.elsevier.com/locate/aescte

Hypersonic vehicle terminal velocity improvement considering ramjet safety boundary constraint

Chengkun Lv [a,1], Zhu Lan [a,2], Tianqi Ma [b,3], Juntao Chang [a,4,*], Daren Yu [a,5]

[a] School of Energy Science and Engineering, Harbin Institute of Technology, Heilongjiang 150001, China
[b] Design Institute of Electro-mechanical Engineering, Beijing 100854, China

A R T I C L E I N F O

Keywords:
Terminal velocity
Hypersonic vehicle
Ramjet
Trajectory optimization
hp adaptive Radau pseudo-spectral method

A B S T R A C T

The air-breathing ramjet is an essential power plant to improve the terminal velocity of hypersonic vehicles. The high-performance requirement of ramjet makes it operate close to the safety boundary, and the optimization of the terminal velocity of vehicle is faced with challenges. An integrated flight/ propulsion model considering thrust and safety performance is constructed, and the trajectory optimization method for the whole flight phase operating process of hypersonic vehicles with multiple performance optimization indicators and complex safety constraints is studied. Moreover, the trajectory optimization results of the whole flight phase for CAV-H and X-43A (powered/unpowered) plans are compared and analyzed, and the dive-acceleration-climb strategy can improve the maneuverability of the vehicle in the ramjet operating phase. Furthermore, the terminal flight velocity of the vehicle in the powered plan is increased from 363.60 m/s to 943.57 m/s under the effect of ramjet. In addition, the trajectory optimization results of vehicle with different geometrical configurations of the ramjet are compared under safety constraints. The results show that the variable geometry ramjet enables the vehicle to increase its terminal velocity by up to 9.73 % with respect to the fixed geometry engine. Hypersonic vehicle terminal velocity is effectively improved by considering ramjet variable geometry adjustment.

Introduction

Since the 21st century, hypersonic vehicles have become the research focus of countries around the world by the advantages of fast flight speed, high maneuverability, and long-range precision strikes [1]. With the continuous improvement of the requirements of modern war missions, hypersonic vehicles are required to perform faster and farther [2]. According to the power plants, hypersonic vehicles can be divided into rocket, ramjet and unpowered glide. Among them, ramjet obtains thrust through the reaction between the oxygen inhaled during flight and the fuel carried by itself. Compared with rocket, ramjet does not need to carry oxidizer, which greatly reduces the overall mass of the vehicle [3]. Ramjet is a reliable power plant to improve the flight velocity and defense penetration ability of hypersonic vehicles and implement long-range precision strike, which has important strategic significance and extremely high application value.

The Hyper-X project began development of the X-43 series of hypersonic vehicle, which cost 230 million dollars and set a new world record with the velocity of Mach 9.8 [4]. In 2013, Lockheed Martin’s Skunk Works announced that it was developing the SR-72 [5], which is powered by TBCC engine and has a flight range of 4800 km and a top velocity of Mach 6. It also has strategic stealth capability and integrates intelligence gathering, reconnaissance, monitoring and strike functions. The Air Force Research Laboratory and the Defense Advanced Research Projects Agency co-hosted the HyTECH project to develop the X-51A [6], which flew for about six minutes at Mach 5.1 and had a flight range of 426 km. In August 2020, Air-Launched Rapid Response Weapon completed its first flight test, and the US Air Force revealed that it flew between Mach 6.5 and 8. The U.S. Army’s Long Range Hypersonic Weapon project continued to advance in 2021, and the Air Force’s Hypersonic Attack Cruise Missile project was scheduled to be announced

* Corresponding author.
E-mail address: changjuntao@hit.edu.cn (J. Chang).
1 School of Energy Science and Engineering
2 School of Energy Science and Engineering
3 Design Institute of Electro-mechanical Engineering
4 Professor, School of Energy Science and Engineering
5 Professor, School of Energy Science and Engineering

https://doi.org/10.1016/j.ast.2023.108804
Received 25 July 2023; Received in revised form 25 October 2023; Accepted 30 November 2023
Available online 5 December 2023

Nomenclature

A	area
A_{th}	nozzle throat area
C_D	lift coefficient
C_L	drag coefficient
D	aerodynamic drag
E	mechanical energy
F	thrust
H	flight altitude
I_{sp}	specific impulse
J	cost function
l	flight range
L	aerodynamic lift
m	vehicle mass
Ma	Mach number
q	dynamic pressure
q_m	mass flow
r	geocentric distance
S	reference area
V	vehicle velocity
α	angle of attack
α_T	thrust deflection angle
γ	flight-path angle
ε	error
ξ	inlet safety margin
ρ	atmospheric density
φ	fuel equivalence ratio

Subscripts

0	initial value
f	terminal value
1–6	engine section

Abbreviation

AoA	angle of attack
LGR	Legendre-Gauss-Radau
NLP	nonlinear programming

and officially launched in 2022 [7]. In March 2018, Russia announced the "Dagger" hypersonic ballistic missile, which is powered by a rocket engine and a ramjet engine, with a flight velocity of Mach 10 and a flight range of more than 2000 km. It is mainly used for precision strikes on land and maritime targets. In July 2018, the "Pioneer" hypersonic boost-glide missile was unveiled, with a maximum flight range of 6000 km and a maximum velocity of Mach 20. It was claimed to be able to break through all air defense and anti-missile systems in the world, which is a milestone for the development of hypersonic weapons from technology research and development to actual combat equipment. In 2020, Russia's "Zircon" hypersonic cruise missile successfully completed four shipborne launch tests, with a flight range of more than 500 km and a maximum flight Mach number of 8 [8]. And in 2021, Russia carried out intensive hypersonic missile launch tests of "Zircon", which was planned to be put into service in 2023.

According to the characteristics of each phase in the flight process of hypersonic vehicle, its flight path can be divided into three parts: ascent, cruise or glide and dive. Typical hypersonic vehicles can be divided into two types: boost-glide vehicles and air-breathing cruise vehicles according to their different working modes [9]. The former is powered during the ascent and cruise phases and unpowered during the dive phase [10], while the latter is powered only during the ascent phase and then separates from the booster to enter the unpowered glide phase [11]. For boost-glide hypersonic vehicles, Dileep et al. [12] studied the trajectory optimization of the ascent phase of a single-stage liquid rocket. Li et al. [13] combined the normal boundary intersection method and pseudo-spectral method for multi-objective trajectory optimization of glide vehicles. Shen et al. [14] investigated the penetration trajectory optimization problem when a hypersonic glide vehicle encountered two interceptor missiles. For air-breathing cruising hypersonic vehicles, He et al. [15] introduced control constraints into trajectory optimization and used an improved pigeon-inspired optimization algorithm to obtain the trajectory of the ascent phase with the least fuel consumption. Yan et al. [16] proposed a penetration trajectory optimization algorithm for an air-breathing hypersonic vehicle to enable the vehicle to avoid intercepting missiles with minimum energy consumption. Yang et al. [17] used Gauss pseudo-spectral method to find the optimal trajectory of the ascent phase of the general hypersonic vehicle considering the flight/propulsion coupling characteristics. Chuang et al. [18] proposed a new method to solve the problem of periodic optimal control, which solved the problem of hypersonic vehicle fuel optimization cycle control. Song et al. [19], aiming at the glide phase of a dual-mode ramjet engine, used Gauss pseudo-spectral method for trajectory optimization and applied hybrid optimal control to engine mode transition. The results showed that hybrid optimal control has a better effect on maximizing the flight range of vehicle. In general, on the premise of completing flight missions, the trajectory optimization of vehicle is to obtain the flight trajectory with the optimal performance under various constraints by establishing the optimization indicator functions and optimizing the variation law of the control variables [20,21]. Considering the nonlinearity and multiple constraints of the optimal problem, numerical methods are good choice at present [22,23]. Numerical methods are mainly categorized into two types: indirect method and direct method [24]. Among them, the indirect method first obtains the first-order necessary conditions of the optimal control problem based on the Pontryagin maximum principle, and then transforms it into the Hamiltonian boundary value problem, and then solves the boundary problem [25]. The derivation and calculation process of the indirect method is very cumbersome and requires an accurate initial value guess for the boundary conditions, so it is not very practical to apply. The direct method first transforms the optimal control problem into a nonlinear programming (NLP) problem by discretizing the state variables and control variables, and then solves it numerically by an NLP solver [26,27]. Although the solution accuracy is not as high as that of indirect method, the cost of direct method is obviously lower within the allowed accuracy range [28]. In addition, the thrust of the engine is often represented by the thrust coefficient in the current studies. Although this method is conducive to the research of tracking control, the internal operating mechanism and safety constraints of engines are ignored, and the changes brought by engine models in different configurations are not considered. At present, the main plans of powered glide are periodic cruise and skip glide, both of which are intermittent ignition. There are few studies on the plan of starting the engine at the end of the glide phase.

In summary, aiming at the weaknesses of the boost-glide vehicle with low terminal velocity and the air-breathing cruise vehicle with short flight range, a powered glide flight plan with a dual-mode ramjet for rapid maneuvering in dive phase is proposed, and the trajectory optimization study of all the flight phases is carried out to analyze the influence on flight performance parameters with or without power. In the trajectory optimization problem, the effect of engine geometry configuration is also considered. This paper is organized as follow: Section 2 builds an integrated flight/propulsion model for trajectory optimization; In Section 3, the whole flight phase trajectory optimization problem is constructed by considering the optimization method, optimization indicators and constraints; Section 4 discusses the trajectory optimization results under different conditions in detail.

Integrated flight/propulsion model

The advantage of the powered flight plan in dive phase is that it can give full play to the performance benefits of the ramjet. In the trajectory optimization of the ramjet operating process, it is very important to describe the operating state of the engine quickly and accurately. In this section, a dynamic model of hypersonic vehicle suitable for boost-glide in the full velocity domain is constructed. Then, the component-level model of a ramjet is constructed, and the flow parameter calculation process of each internal section is given. The performance characterization parameters such as thrust and inlet safety margin are also calculated under the corresponding modes. Combined with the characteristics of the ramjet engine and the dynamic model of the vehicle, an integrated flight/propulsion model for trajectory optimization is constructed.

Vehicle modeling

The dynamic model of vehicles can be expressed as ref. [21]:

$$\begin{cases} \dot{r} = V\sin\gamma \\ \dot{V} = (F\cos\alpha - D)/m - g\sin\gamma \\ \dot{\gamma} = (F\sin\alpha + L)/(mV) + (V/r - g/V)\cos\gamma \\ \dot{m} = -F/\left(gI_{sp}\right) \\ \dot{l} = R_0 V\cos\gamma/r \\ \dot{\alpha} = k_1 \\ \dot{\phi} = k_2 \end{cases} \tag{1}$$

where, r is the distance from the vehicle to the earth center, V is the vehicle velocity, γ is the flight-path angle, m is the vehicle mass, and l is the flight range. The control variables in the actual model are angle of attack, α, and ramjet fuel equivalence ratio, φ, respectively. Meanwhile, to prevent high-frequency oscillations in the values of angle of attack and fuel equivalent ratio during the optimization process, they are used as state variables and their rate of change is limited. F and I_{sp} are thrust and specific impulse of the engine respectively. R_0 is the average radius of the earth, whose value is 6,378,145 m. The flight altitude of the vehicle above the ground $H = r\text{-}R_0$. g is the acceleration of gravity at the altitude H, its calculation formula is $g = \mu/r^2$. And μ is the gravitational coefficient of the earth, its value is 3.986×10^{14}. L and D are the aerodynamic lift and aerodynamic drag respectively, and their calculation formulas are shown in Eq. (2). C_L and C_D are the lift and drag coefficients respectively. q is the dynamic pressure. S is the aerodynamic reference area of the vehicle. ρ is the atmospheric density, and the atmospheric environment parameters are calculated using the standard atmospheric model defined in the United States in 1976.

$$\begin{cases} L = C_L qS \\ D = C_D qS \\ q = 0.5\rho V^2 \end{cases} \tag{2}$$

The X-43A vehicle model considers the variation of aerodynamic coefficients over a wide range of angles of attack and Mach numbers, and is widely used in the field of trajectory optimization for air-breathing hypersonic vehicles. According to the Mach number and angle of attack intervals of each phase, the reference data are censored and rectified into two parts for fitting. The first part of the lift and drag coefficients corresponds to the Mach number interval of 2–15 and the angle of attack interval of $0^\circ - 20^\circ$, which is used for the inertial ascent and reentry glide phases. And the second part of the lift and drag coefficients corresponds to the Mach number interval of 1–8 and the angle of attack interval of $-5^\circ - 10^\circ$, which is used for the ramjet and dive phases.

The fitting models of lift coefficient C_L and drag coefficient C_D in each phase with respect to angle of attack and Mach number are obtained based on the least square identification method. The fitted models and reference data with Mach number are shown in Fig. 1 and Fig. 2. It can be seen that the model can fit data smoothly and accurately, and the fitting effect is satisfactory.

The boost-glide vehicle studied in this paper generally has a total flight range of about 4000km~5000 km. According to ref. [29], a two-stage solid rocket is used as the power plant in the rocket booster phase, and the specific parameters are shown in the following Table 1. Meanwhile, it is assumed that the angle of attack is small during the flight of this phase and the aerodynamic lift is approximately 0. Therefore, the lift coefficient of the rocket booster phase is set to be constant at 0, the drag coefficient is constant at 0.5, and the aerodynamic reference area $S = 3m^2$. The flight trajectory is optimized by adjusting the thrust deflection angle, α_T, of the rocket.

Ramjet modeling

The component-level model of a ramjet is constructed according to its operating mechanism, which is a type of model with complex thermal variations and strong nonlinearity, and mainly includes three components: inlet, combustor and nozzle. The component-level modeling process is simply described as follows: first, the Mach number and altitude of the current flight conditions are used to calculate the outflow parameters in the atmospheric model, and the incoming flow parameters and angle of attack are used as inputs to the inlet, while the fuel equivalence ratio is used as input to the combustor, and the performance parameters such as thrust and margin are finally obtained through the flow parameters of each component. The model controls the fuel equivalence ratio and angle of attack to influence the engine operating conditions and thus change the performance parameters.

The flow in the inlet can be expressed by the isentropic equation as follows:

Fig. 3.

$$\begin{cases} \dot{m}_i = KA_i\dfrac{p_i^*}{\sqrt{T_i^*}}q(Ma_i) = const \\ T_i^* = const \\ p_i^* = const \end{cases} \quad i = 1, 2 \tag{3}$$

Assuming that the calorific value of fuel in the combustor is Q and the combustion efficiency is η=0.99. According to the conservation of energy, the energy equation before and after adding fuel can be obtained as Eq. (4). In the equation, f_{st} is the stoichiometric ratio of fuel and air, c_p is the specific heat, φ is the fuel equivalent ratio, which is greater than 1 means excess fuel, less than 1 means excess air, and τ is the total temperature rise ratio of the combustor.

$$q_{m3}T_3^*c_p + q_{m3}Q\eta f_{st}\varphi = q_{m4}T_3^*c_p(1+f_{st}\varphi)\tau \tag{4}$$

The flow in the combustor can be regarded as equal cross-sectional heat transfer process [30], in which the total temperature rise ratio function is introduced, as shown in Eq. (5):

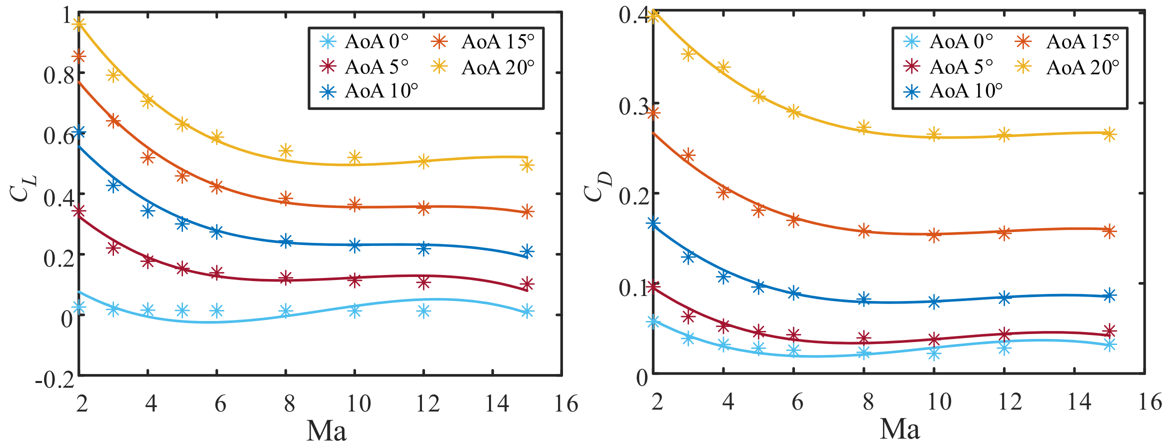
$$\tau = \left[\frac{\Phi(Ma_4)}{\Phi(Ma_3)}\right]^2 \tag{5}$$

where the total temperature rise ratio function is expressed as follows:

$$\Phi(Ma) = \frac{Ma\left[1 + \left(\frac{k-1}{2}\right)Ma^2\right]^{\frac{1}{2}}}{1 + kMa^2} \tag{6}$$

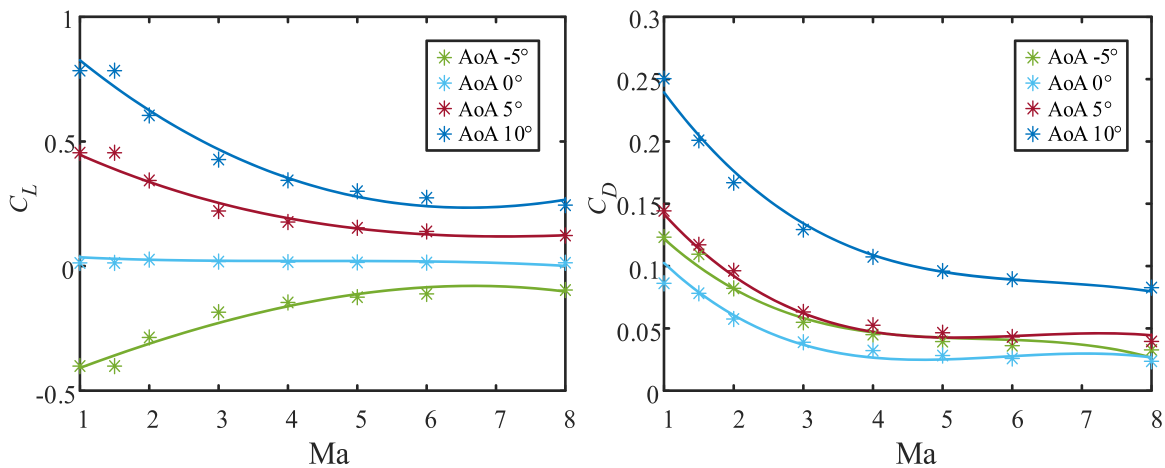
The nozzle equation is the same as the inlet equation:

$$\begin{cases} \dot{m}_i = KA_i\dfrac{p_i^*}{\sqrt{T_i^*}}q(Ma_i) = const \\ T_i^* = const \\ p_i^* = const \end{cases} \quad i = 4, 5, 6 \tag{7}$$



(a) Aerodynamic lift coefficient (b) Aerodynamic drag coefficient

Fig. 1. The first part of the lift and drag coefficients (points represent the aerodynamic reference data of X-43A vehicle at different angles of attack and lines represent the fitting model).



(a) Aerodynamic lift coefficient (b) Aerodynamic drag coefficient

Fig. 2. The second part of the lift and drag coefficients (points represent the aerodynamic reference data of X-43A vehicle at different angles of attack and lines represent the fitting model).

Table 1
Mass parameters and flight loads of two-stage booster rocket.

Parameter	First-stage	Second-stage
Rocket fuel mass [kg]	12,163	5085
Total rocket mass [kg]	14,309	5650
Thrust [kN]	420.6	140.1
CAV-H mass [kg]	907.2	
X43-A mass + fuel mass [kg]	1280.6 + 150	

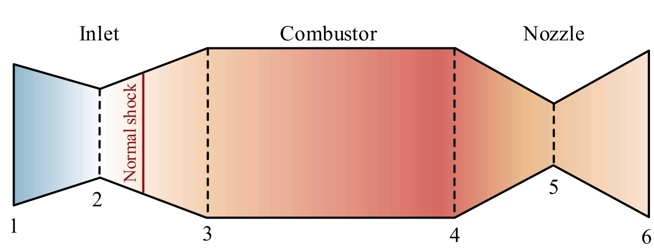


Fig. 3. Schematic diagram of ramjet engine.

The inlet safety margin ξ is an important parameter to reflect whether the operating state of the ramjet is safe, but it cannot be measured directly. Therefore, referring to the calculation method ref. [31], its calculation formula is given in Eq. (8). Firstly, the maximum and minimum static pressure of the inlet outlet, $p_{3\max}$ and $p_{3\min}$, can be calculated according to the operating conditions and geometric conditions of the engine. Then the static pressure p_3 of the inlet outlet under the actual operating state is calculated iteratively, and the inlet safety margin is obtained. When the margin value is less than 0, the engine is in unstart state; when the margin value is greater than 0 and less than 1, the engine is in ramjet mode; when the margin value is greater than 1, the engine is in scramjet mode.

$$\xi = \frac{p_{3\max} - p_3}{p_{3\max} - p_{3\min}} \tag{8}$$

So far, the Mach number, total temperature, total pressure and other flow parameters of each section can be calculated. And the engine thrust is calculated as follows.

$$F = \dot{m}_6(V_6 - V_0) + (p_6 - p_0)A_6 \tag{9}$$

Trajectory optimization oriented integrated flight/propulsion model

The control variables of integrated flight/propulsion model used in this study are $U = [\alpha, \varphi, A_{th}]$. The input parameters of the ramjet model include Mach number, altitude, nozzle throat area, angle of attack and fuel equivalence ratio. The output performance parameters are thrust and fuel mass flow, and the output safety parameter is inlet safety margin. Angle of attack, thrust, and fuel mass flow are used as inputs to the vehicle model to adjust its state in real time, while the real-time altitude and Mach number of the vehicle are fed back to the engine, as shown in Fig. 4.

The engine optimization model not only requires that the structural form should not be too complex, but also can obtain the optimal solution for the integrated optimization model, and can accurately obtain the performance parameters and safety parameters of the engine under a wide range of flight conditions. The component-level model has high precision, but it is difficult to be solved quickly during trajectory optimization. Therefore, it is necessary to fit a multi-input multi-output engine model for trajectory optimization based on the characteristic laws of the component-level model. A simple and reliable engine optimization model is constructed as shown in Eq. (10) and (11). The fitting results with different geometric configurations are shown in Fig. 5.

$$F = \sum_{i=0}^{1} \sum_{j=0}^{2-i} k_{Fij} Ma^i \dot{m}_{fuel}^j \tag{10}$$

$$\xi = \sum_{i=0}^{4} \sum_{j=0}^{3-i} k_{\xi ij} Ma^i \varphi^j \tag{11}$$

Trajectory optimization

Trajectory optimization method

In essence, the trajectory optimization of a vehicle can be abstracted as a class of continuous-time optimal control problems with strong nonlinearity for solving the extreme values of performance optimization indicator functions under various constraints such as kinetic differential equations, algebraic equations and inequality equations. In this study, Radau pseudo-spectral method and hp adaptive strategy are used to solve the trajectory optimization problem.

The Radau pseudo-spectral method, as one of the direct methods in the numerical solution, firstly performs a time domain transformation, and selects Legendre-Gauss-Radau (LGR) collocation points in the interval $[-1,1]$ for discretization. Then, Lagrange global interpolation polynomials are constructed to approximate the state and control variables discretized at the collocation points, and the interpolation polynomials are derived to transform the original dynamic differential equation constraints into a set of algebraic equation constraints. Finally, the discretized state and control variables are introduced in the terminal constraints, process constraints, and other constraints to discretize them. The Gauss-Legendre product formulation is used to approximate the integral term in the performance optimization indicator functions, thus transforming the continuous-time optimal control problem into an NLP problem and solving it [32,33].

The hp adaptive strategy evaluates the accuracy of each interval during each iteration. When the calculation error of state variables and path constraints between any collocation points in the interval is greater than the maximum allowable error, the interval needs to be refined. The choice of refining the number of intervals or raising the order of interpolation polynomials depends on the calculation error magnitude of the state variables and path constraints between the two collocation points in the interval. If the error magnitude is comparable between all the collocation points in the interval, the calculation accuracy is improved by raising the interpolating polynomial order; if the error magnitude of one point in the interval is larger than that of the other points, this interval should be refined [34,35].

Optimization problem construction

Before constructing the trajectory optimization problem, each flight phase of the vehicle should be defined systematically. The flight plan considering an unpowered vehicle is generally divided into: rocket booster phase, inertial ascent phase, reentry glide phase and dive phase. The abbreviations for each flight phase are shown in Table 2. On the basis of the flight plan with unpowered vehicle, the ramjet engine ignites and starts after the powered vehicle glides to the designated point. Then the maneuvering flight is carried out according to the given performance optimization indicators to complete the flight mission within the engine safety performance constrained. After reaching the designated position with the fuel depleted, the final dive is performed, as shown in Fig. 6.

Control variables and optimization indicators

The control variables and optimization indicators for each phase are shown in Table 3.

Constraints

Constraint is an indispensable factor in optimal control problems, especially in multi-phase trajectory optimization. It is helpful to construct a solvable trajectory optimization problem by setting the constraint conditions of each phase reasonably.

In order to ensure the smooth development of the flight plan, many constraints in addition to dynamic differential equation constraints must be met in each phase: 1) Variable constraints: including the control variables and state variables of the model in each phase; 2) Process constraints: including bending moment constraints for booster rocket, dynamic pressure, overload and heat flux constraints for vehicle safety, inlet safety margin constraint for engine safety; 3) Terminal constraints: including the beginning and end states of each phase; 4) Event constraints: connecting each phase to maintain parameter continuity.

Variable constraints. The thrust deflection angle, α_T, in the RB-phase

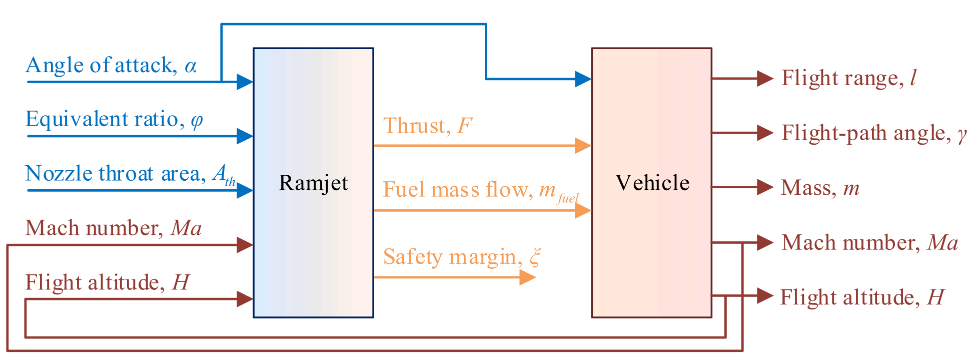


Fig. 4. Schematic diagram of integrated flight/propulsion model.

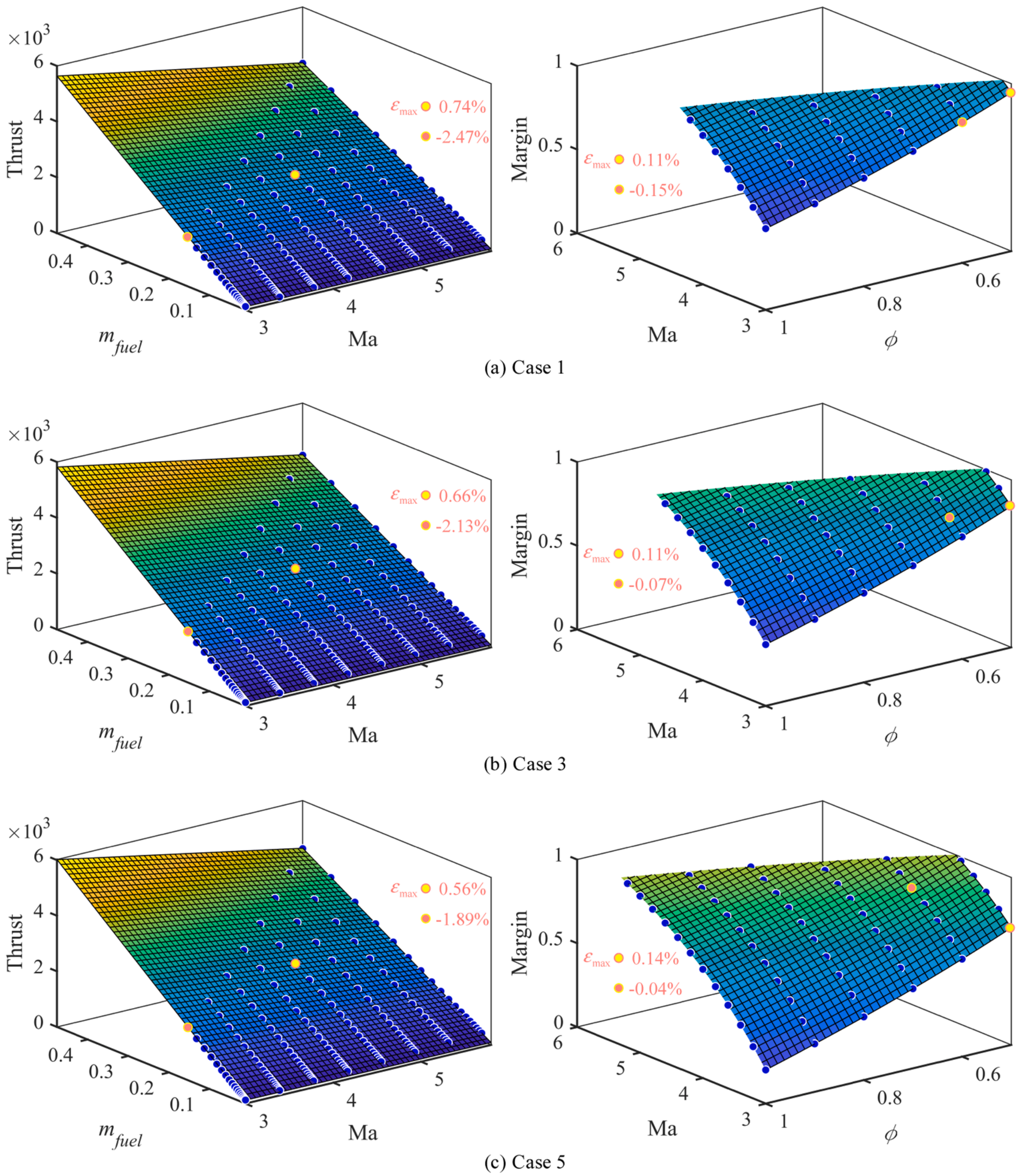


Fig. 5. Fitting results for F and ξ in different cases. The nozzle throat area is 0.039 m^2 in Case 1; the nozzle throat area is 0.035 m^2 in Case 3; the nozzle throat area is 0.031 m^2 in Case 5.

Table 2
Abbreviation for each phase.

Flight phase	Abbreviation
Rocket booster phase	RB-phase
Inertial ascent phase	AS-phase
Reentry glide phase	GL-phase
Ramjet phase	RA-phase
Dive phase	DI-phase

should be reasonably constrained, the angle of attack in each phase and the fuel equivalence ratio in the RA-phase as the actual control variables should also be constrained.

$$\begin{aligned} &\alpha_{T\min} \leq \alpha_T \leq \alpha_{T\max} \\ &\alpha_{\min}^{(i)} \leq \alpha^{(i)} \leq \alpha_{\max}^{(i)} \\ &\phi_{\min} \leq \phi \leq \phi_{\max} \end{aligned} \tag{12}$$

Where i represents the i phase, $i = 1, 2, 3, 4, 5$.

As the control variables of the model, the change rate of angle of attack and the change rate of fuel equivalence ratio also need to be constrained.

$$\begin{aligned} &k_{1\min} \leq k_1 \leq k_{1\max} \\ &k_{2\min} \leq k_2 \leq k_{2\max} \end{aligned} \tag{13}$$

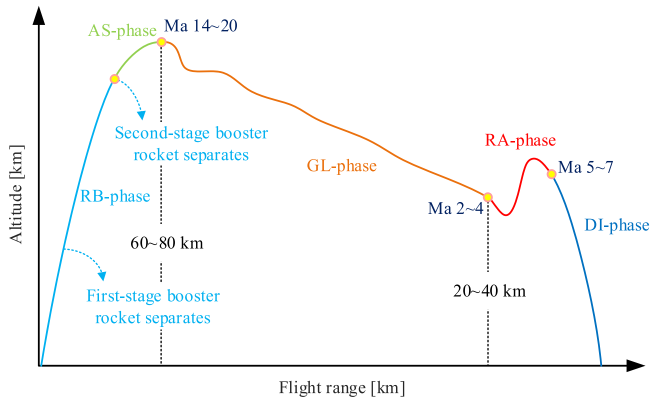


Fig. 6. Schematic diagram of the whole flight phases of the powered flight plan.

Table 3
Control variables and optimization indicators for each phase.

Flight phase	Control variables	Optimization indicators
RB-phase	Thrust deflection angle α_T	min J=-E_f
AS-phase	Angle of attack α	min J=-l_f
GL-phase	Angle of attack α	min J=-l_f
RA-phase	**Angle of attack α**	**min J=-V_f**
	Fuel equivalent ratio φ	
	Nozzle throat area A_{th}	
DI-phase	Angle of attack α	min J=-V_f

In order to ensure that the vehicle operates in reasonable conditions at each phase, the state variables need to be constrained.

$$\begin{aligned} & H_{\min}^{(i)} \le H^{(i)} \le H_{\max}^{(i)} \\ & V_{\min}^{(i)} \le V^{(i)} \le V_{\max}^{(i)} \\ & \gamma_{\min}^{(i)} \le \gamma^{(i)} \le \gamma_{\max}^{(i)} \\ & m_{\min}^{(i)} \le m^{(i)} \le m_{\max}^{(i)} \\ & l_{\min}^{(i)} \le l^{(i)} \le l_{\max}^{(i)} \end{aligned} \tag{14}$$

Process constraints.

(1) RB-phase

Bending moment constraint: there is a problem of bending moments generated by excessive aerodynamic loads during the flight of rockets through dense atmospheres [36]. The bending moment should be constrained in order to avoid excessive thrust deflection angles in the high dynamic pressure region that can be harmful to the structural safety of the rocket. The bending moment is calculated as shown in Eq. (15). M is the bending moment, and $M_{\max}$ is the set maximum value of the bending moment.

$$M = |q\alpha_T| \le M_{\max} \tag{15}$$

(1) AS-phase and GL-phase

In the AS-phase and the initial GL-phase, the vehicle has the highest velocity in the whole process. At this time, the mechanical energy of the vehicle is the maximum, and the dynamic pressure, heat flux and overload increase sharply, which has adverse effects on the vehicle. Three process constraints are mathematically described.

Dynamic pressure constraint: The large moment caused by excessive dynamic pressure will exceed the carrying capacity of the vehicle's rudder surface execution mechanism, which in turn will have an adverse impact on mechanical structure of the vehicle. Structural failure will lead to mission failure, so the dynamic pressure needs to be limited, and the calculation formula is shown in Eq. (16), $q_{\max}$ is the set maximum value of dynamic pressure.

$$q = \frac{1}{2}\rho V^2 \le q_{\max} \tag{16}$$

Heat flux constraint: It should be noted that the high temperature will ablate the surface of the vehicle and seriously threaten the structural safety of the vehicle. Therefore, the heat flux at the stagnation point is used as a constraint, and the calculation formula is shown in Eq. (17), where K_Q is the heat transfer coefficient related to the radius of the vehicle head. In this study, K_Q =7.9686×10^{-5} Js^2/(m$^{3.5}$kg$^{0.5}$), n is generally taken as 0.5, m is taken as 3, and $\dot{Q}_{\max}$ is the set heat flux constraint value.

$$\dot{Q} = K_Q \rho^n V^m \le \dot{Q}_{\max} \tag{17}$$

Overload constraint: During high-velocity flight, the entire vehicle will be subjected to aerodynamic overload, and too large overload will pose a threat to the structural strength of materials and the normal operation of supporting equipment. This study sets a maximum threshold for total overload, which is calculated as follows.

$$n_L = \frac{\sqrt{L^2 + D^2}}{mg_0} \le n_{L\max} \tag{18}$$

(1) RA-phase

The process constraints for the vehicle in the RA-phase are the same as those for the GL-phase, so the process constraints established for the engine are introduced here. There is a safety boundary for ramjet engines operating in the wide speed domain. Once this boundary is crossed, it is likely to cause a series of serious problems such as engine unstart, loss of thrust and internal structural damage, etc. Therefore, the parameters characterizing the safety performance should be constrained by adjusting the control variables during the trajectory optimization process to ensure that the engine is in proper operating condition.

Inlet safety margin constraint: Ensuring that the inlet is in the starting state is crucial to the successful completion of the flight mission. Characteristic analysis shows that at high Mach number, due to the enhancement of anti-backpressure ability, the margin value is large and there is no phenomenon of inlet unstart. At low Mach number, the pressure in the combustor increases with the increase of equivalence ratio, and the shock wave constantly moves forward in the isolator under the action of back pressure. When the back pressure increases to the point where the shock wave is just pushed out of the isolator, the safety margin is 0, which indicates the inlet unstart. Therefore, limiting the safety margin at low Mach number is an important control objective to ensure the normal operation of the engine. As shown in Eq. (19), $\xi_{\min}$ is the lowest margin value, and a certain margin is left for the safety margin to prevent uncertain factors from causing engine safety problems.

$$\xi_{\min} \le \xi \tag{19}$$

(1) DI-phase

In the process of dive, with the sharp increase of air density, high-velocity flight will lead to a sharp rise in dynamic pressure and overload. However, in order to strengthen the penetration capability of the end, the performance optimization indicator of the DI-phase is the maximum terminal velocity. If some constraints are imposed, the terminal velocity will decrease significantly and it is difficult to achieve the expected effect. Moreover, considering the short duration of the DI-phase, the process constraint is not applied.

Terminal constraints. The constraint range is set according to the altitude, velocity and flight-path angle of the terminals of different flight phases, where $H_f^{(i)}, V_f^{(i)}, \gamma_f^{(i)}$ is the terminal constraint of the ith phase, and $\varepsilon_H^{(i)}, \varepsilon_V^{(i)}, \varepsilon_\gamma^{(i)}$ is the artificially set terminal constraint error of the ith phase, which can be a constant not less than 0. If requirements for precision guidance need to be met, they can all be set to 0. It is important to note that not every phase requires terminal constraints for all three states. The terminal constraint for each phase is determined by the mission.

$$\begin{aligned} &\| H^{(i)}(t_f) - H_f^{(i)} \| \le \varepsilon_H^{(i)} \\ &\| V^{(i)}(t_f) - V_f^{(i)} \| \le \varepsilon_V^{(i)} \\ &\| \gamma^{(i)}(t_f) - \gamma_f^{(i)} \| \le \varepsilon_\gamma^{(i)} \end{aligned} \tag{20}$$

Event constraints. To ensure the continuity of state variables and control variables in the whole phase, event constraints should be established so that the state variables, control variables and time are consistent at the connections of adjacent phases.

$$\begin{aligned} t_f^{(i)} &= t_0^{(i+1)} \\ x_f^{(i)} &= x_0^{(i+1)} \\ u_f^{(i)} &= u_0^{(i+1)} \end{aligned} \tag{21}$$

Optimization process

On the premise of ensuring the smooth connection between two adjacent phases and considering the complex ballistic situation of the whole flight phase, the segmented multi-objective optimization method is adopted. The specific flow chart of optimization is shown in Fig. 7. By determining the control variables, performance optimization indicators and various constraints such as variable constraints, process constraints and terminal constraints for each phase, the trajectory optimization model of each phase is constructed. Then the event constraints are used to connect each phase to complete the establishment of the whole flight phase trajectory optimization model.

After that, the variables are discretized by pseudo-spectral method. The optimization problem is transformed into NLP problem, and then

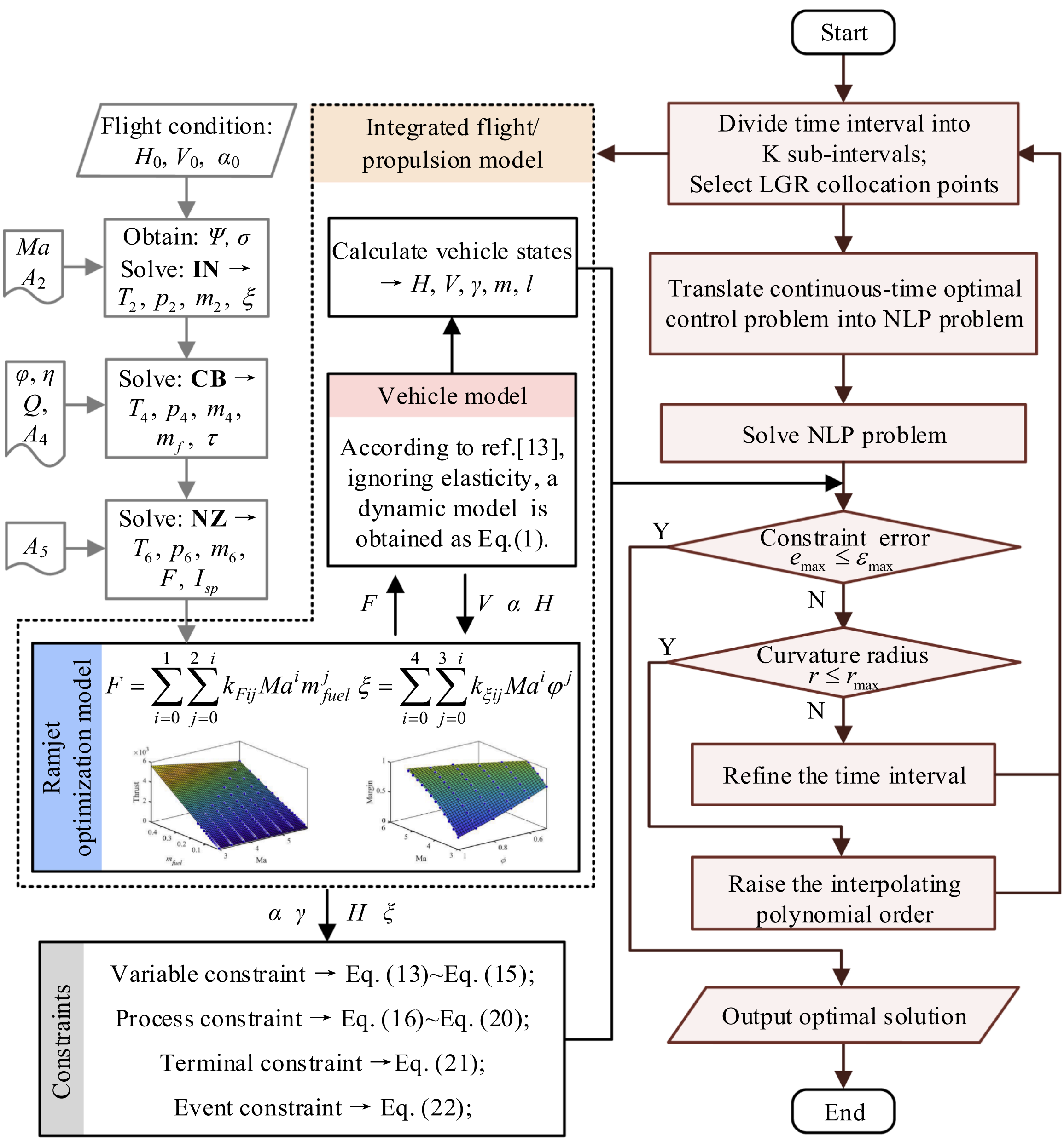


Fig. 7. Flow chart of optimization process.

solved under the number of initial iterative intervals. The constraint function errors of each interval are calculated, and if all intervals meet the accuracy requirements, the solution result is output; if not, the solution within the interval needs to be improved. According to the hp adaptive strategy [34], the curvature radius $r_{\max}$, which characterizes the restriction on the drastic degree of trajectory variation within the interval, is defined to determine the appropriate improvement strategy. If the current curvature radius $r > r_{\max}$, it indicates that the trajectory variation within the interval is too drastic and the number of intervals needs to be increased to meet the accuracy requirement. If the current curvature radius $r \leq r_{\max}$, it indicates that the trajectory variation within the interval is relatively uniform and an improvement strategy of raising the interpolating polynomial order can be adopted.

Results and discussion

In this section, the superiority of the dive-accelerate-climb strategy is discussed in terms of the ramjet operating process in RA-phase. Then, the results of trajectory optimization in powered and unpowered plans are compared and analyzed for the whole flight phase. Finally, the effect of variable geometry ramjet engine on the trajectory optimization results of the RA-phase is discussed for the powered plan.

RA-phase analysis

In RA-phase, the ramjet's mission is to accelerate and drive the vehicle to a specified altitude. The acceleration state implies high fuel consumption, so the ramjet engine needs to rapidly increase the altitude before running out of fuel and to increase the terminal velocity as much as possible to further enhance the defense penetration and attack capability of the DI-phase. In the optimization process of RA-phase, the flight trajectory of the ramjet always shows a dive-accelerate-climb strategy when the flight altitude of the vehicle is not strictly constrained as a state variable, and a more in-depth study is conducted to

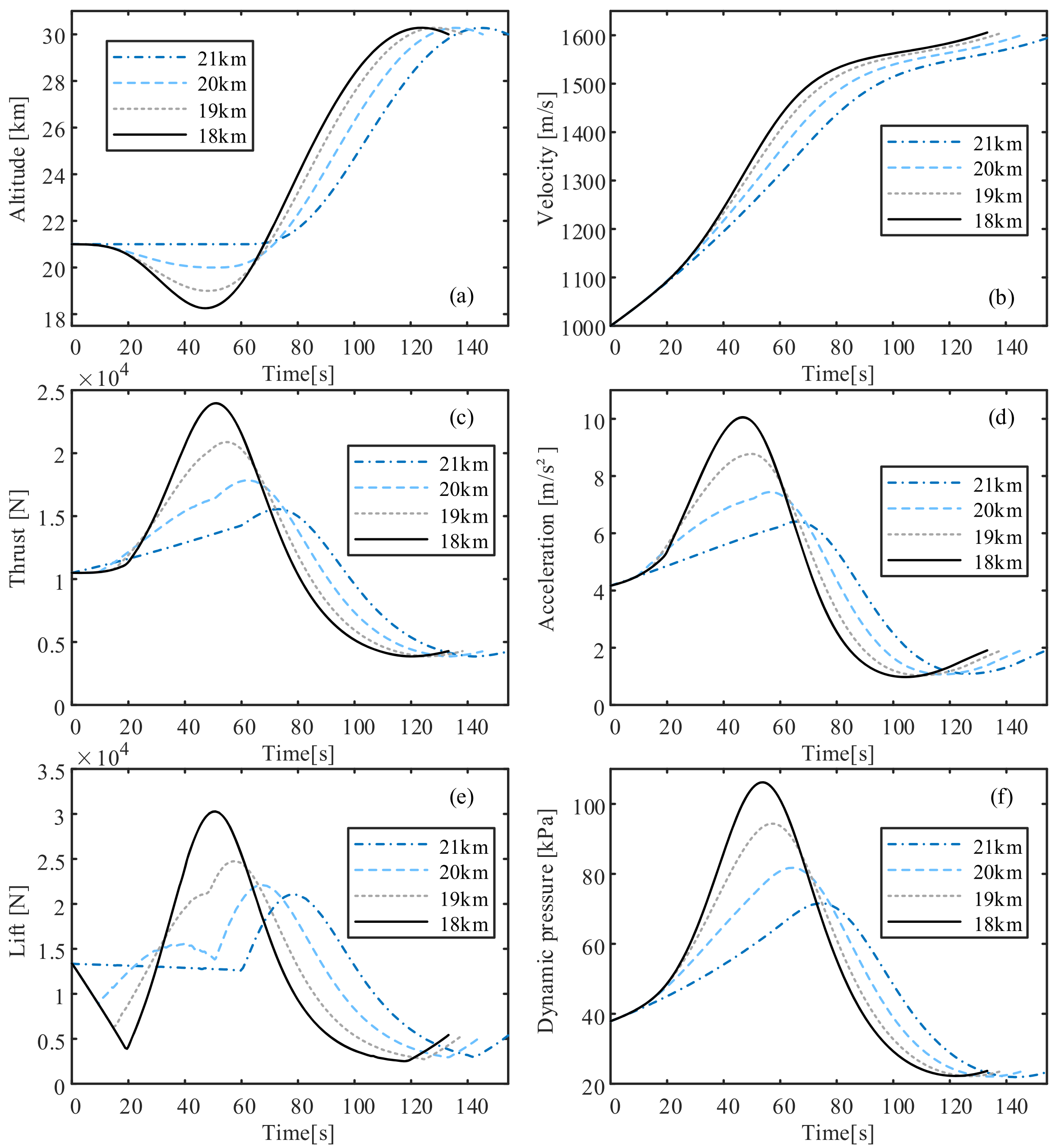


Fig. 8. Optimization results for RA-phase with different lower limits for altitude constraint. The line of 21 km indicates a horizontal acceleration strategy; the lines of 20 km and 19 km indicate limited dive acceleration strategies; and the line of 18 km indicates an unlimited dive acceleration strategy.

address this phenomenon.

A fixed geometry engine with a nozzle throat area of 0.035 m^2 is first considered. In the trajectory optimization process, the lower limit of the vehicle flight altitude constraint is gradually reduced from the initial flight altitude until the flight trajectory in the optimization result is no longer affected by the altitude constraint. Fig. 8 shows these optimization results with different lower limits for altitude constraint. During the ramjet dive, the potential energy of the vehicle and the chemical energy of the fuel contribute to the increase of the kinetic energy of the vehicle, which makes the acceleration in the dive process shown in Fig. 8(d) increase rapidly compared with the horizontal acceleration process. And this increase becomes more and more significant with the decrease of the lower limit. The lower altitude and higher velocity can rapidly increase the dynamic pressure during the flight. By comparing the dynamic pressure change process under different constraints, it can be found that the process of diving is also the process of dynamic pressure raising, and the increase of dynamic pressure helps the accumulation of aerodynamic lift. After diving to the appropriate position, the vehicle already has a high enough aerodynamic lift and thrust, so that under the combined action of both, the vehicle quickly climbs to the target altitude.

Since the starting flight altitude is 21 km, it is a horizontal acceleration strategy when the lower limit of the altitude constraint is set to 21 km. As the altitude constraint decreases, the trajectory optimization result of RA-phase presents the dive acceleration strategy. When the lower altitude constraint is 20 km or 19 km, the lowest altitude of the trajectory in the optimization result is the lower altitude constraint, which indicates that the dive acceleration strategy is limited and not optimal. When the lower altitude constraint is 18 km, the lowest altitude of the trajectory in the optimization result has not reached the limiting value, which means that the dive acceleration strategy is an unlimited optimal solution. By comparing the velocity curves of the unlimited dive acceleration strategy and the horizontal acceleration strategy, it is found that the terminal velocity of the dive-accelerate-climb strategy is increased by 0.74 % for the same total fuel consumption. Although this difference is not significant, the comparison of the altitude change process shown in Fig. 8(a) reveals that the dive-accelerate-climb strategy takes less time to reach the same altitude, and saves 21.16 s, which further improves the vehicle's defense penetration capability. The reason for this gap is that the vehicle achieves a rapid increase in velocity, thrust and aerodynamic lift through the dive process. Compared with the strategy of horizontal acceleration before climbing, the maximum thrust and aerodynamic lift of the former in the RA-phase are increased by 54.05 % and 43.76 % respectively. Moreover, the maximum aerodynamic lift in the dive-accelerate-climb strategy is increased by 126.14 % compared with the initial moment. This not only shows the superiority of the dive-accelerate-climb strategy, but also illustrates the influence of the boundary conditions on the trajectory optimization results.

Trajectory optimization results

In the unpowered flight plan, the vehicle is first sent to a certain altitude through a two-stage solid rocket in the RB-phase. And at the end of the boost, the inertia is used to further lift the vehicle to the maximum flight altitude. The unpowered reentry glide is then performed, and after gliding to the designated point, the vehicle performs a dive to obtain maximum kinetic energy to destroy the target. In the powered plan, the ramjet engine will start at the end of the GL-phase and stop working after completing the assigned mission, and the vehicle will enter the DI-phase. The three flight plans considered in this study are shown in Table 4.

Table 4
Flight plans.

Plan	Vehicle	Ramjet power
A	CAV-H	unpowered
B	X-43A	unpowered
C	X-43A	powered

Fig. 9 shows the optimization results for all phases in different plans. The operating trajectory of GL-phase usually adopts skip-glide type, which has longer flight range and greater maneuverability than equilibrium glide trajectory. The reason for the formation of skip-glide trajectory is that the lift of the vehicle in the initial moment of gliding is less than its own gravity, the vehicle is first diving gliding. With the decrease in altitude, the air density increases and the dynamic pressure gradually increases, resulting in a gradual increase in lift until greater than gravity, the vehicle turns to climb gliding. As the flight altitude increases, the air density decreases, leading to a decrease in dynamic pressure, and the lift gradually decreases until it is less than the gravity of the vehicle. A skip-glide trajectory is formed as a result of such a reciprocating process. By the influence of aerodynamic drag, the kinetic energy of the system will continue to reduce, the skip amplitude will show a decreasing trend, as shown in Fig. 9(a). It is worth noting that the skip amplitude of the X-43A vehicle is smaller during the glide phase because its larger aerodynamic reference area results in a smaller dynamic pressure condition for the lift to balance with gravity in the vertical direction. This means that the X-43A vehicle can make repeated skips at lower air density and higher altitude conditions.

The flight range, altitude and velocity distributions of the three plans are shown in Table 5. Plan A, Plan B and Plan C have a total flight range of 5329.26 km, 4420.43 km and 4338.44 km, respectively. Compared with Plan B, the total flight range of CAV-H, an unpowered gliding hypersonic vehicle with a high lift-to-drag ratio, is increased by 908.83 km, which is an increase of 20.56 %. For X-43A vehicle, the total flight range of Plan B is only slightly higher than that of Plan C. In these optimization results, the effect of the presence of power on the total flight range of the vehicle is not significant. In addition, the impact velocities of the three at the end of the DI-phase reaches 341.06 m/s, 363.60 m/s and 943.57 m/s, respectively. Among them, the terminal impact velocity of the powered Plan C is significantly higher than that of the unpowered Plan A and Plan B, and this increase reaches 159.51 %. In contrast to the long flight range demonstrated in Plan A, the difference in vehicles has no significant effect on the terminal impact velocity.

Obviously, the difference in terminal performance of different plans is the result of the combined effect of each phase. In RB-phase, since the goal is to increase the mechanical energy of the vehicle as much as possible, the flight range of the three plans in this phase is not very different, and the proportions relative to the total flight range are only 5.10 %, 5.95 % and 5.91 %. Among them, RB-phase is divided into a first-stage booster phase and a second-stage booster phase. The initial conditions for each booster phase in the RB-phase have been shown in Table 1. Although the thrust of the second-stage booster rocket is much smaller than that of the first-stage booster rocket, the altitude increment of vehicle boosted by the second-stage booster rocket is larger than that of the first-stage booster rocket due to the impact of the flight load, and the velocity increase has also reached twice as much. The velocities of the three plans at the end of the RB-phase are 4723.16 m/s, 4482.64 m/s, and 4315.10 m/s, respectively. Because of the lower mass of the CAV-H vehicle, the velocity of Plan A at the end of the RB-phase is slightly higher than that of Plan B, and due to the quality of fuel, the velocity of Plan B is slightly higher than that of Plan C. For AS-phase, the flight ranges of the three plans are 240.97 km, 226.56 km and 200.16 km, accounting for 4.52 %, 5.13 % and 4.61 % of the total flight range respectively. In addition, affected by the initial velocity of AS-phase, Plan A always maintains a higher flight velocity in this phase. After reaching the maximum altitude of the entire flight, vehicles enter the GL-phase and perform an unpowered glide with a skip-glide trajectory. The optimization indicator of this phase is the longest flight range, and the flight range of each plan is 4706.16 km, 3880.42 km and 3597.75 km, which is mainly due to the initial velocity of GL-phase the aerodynamic shape of the vehicle. The initial glide velocity of each plan is 4703.01 m/s, 4449.94 m/s and 4286.82 m/s, respectively. As can be

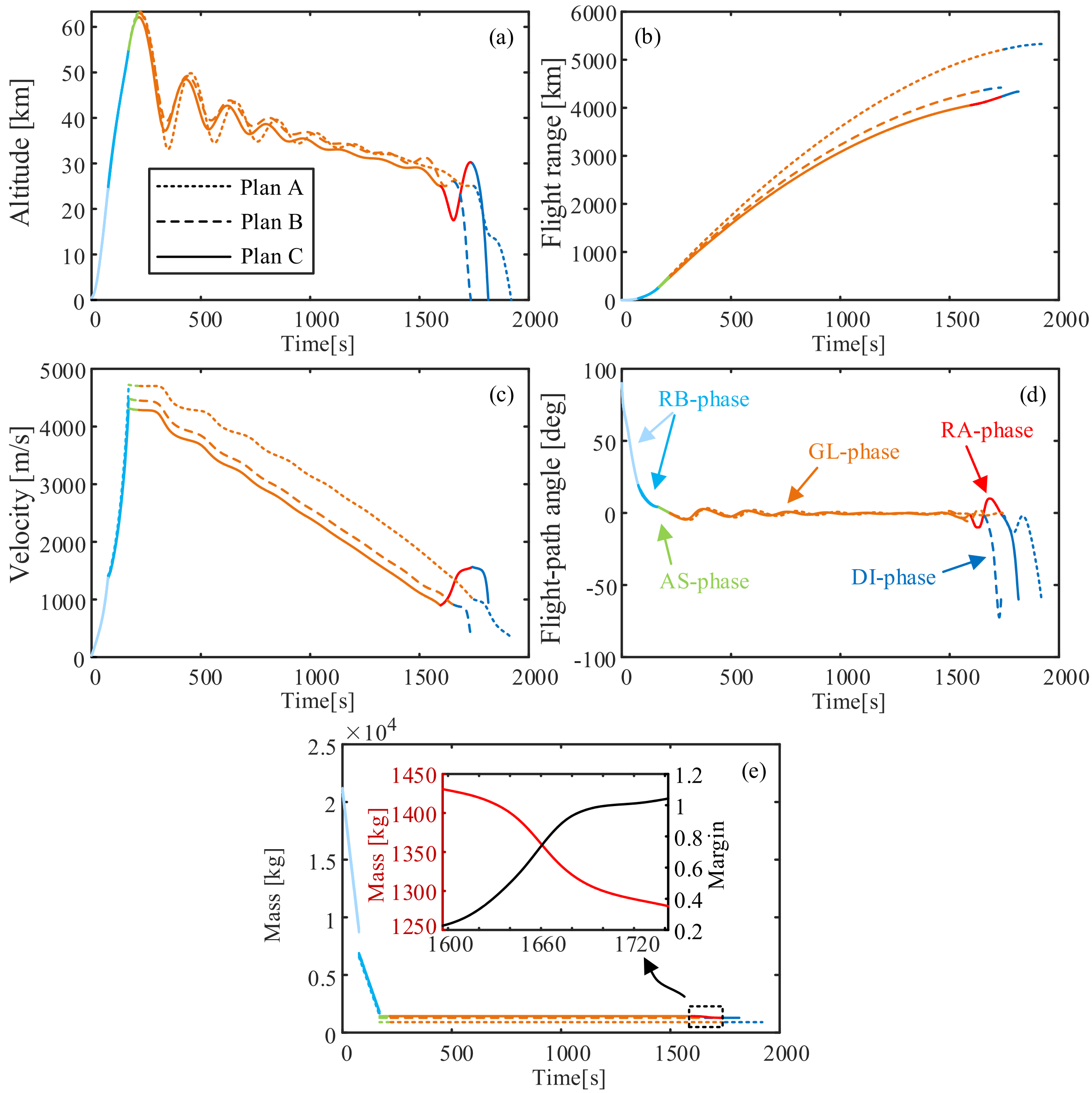


Fig. 9. Optimization results for all phases in different plans.

Table 5
The terminal states of each phase in different plans.

Terminal	Plan	RB-phase	AS-phase	GL-phase	RA-phase	DI-phase
H [km]	A	55	63.33	24.99	–	0
	B	55	62.98	25.00	–	0
	C	55	62.11	25.00	30	0
V [m/s]	A	4723.16	4703.01	999.64	–	341.06
	B	4482.64	4449.94	905.46	–	363.60
	C	4315.10	4286.82	899.76	1563.89	943.57
L [km]	A	271.55	512.52	5218.68	–	5329.26
	B	262.86	489.42	4370.28	–	4420.43
	C	256.50	456.66	4054.41	4240.93	4338.44

seen from Fig. 9(c), if the initial velocity of GL-phase is small, its velocity is the smallest at any moment of the full GL-phase, and the terminal point of the GL-phase is reached earlier. In each plan, the GL-phase accounts for 88.31 %, 87.79 % and 82.93 % of the total flight range respectively. It is obvious that the GL-phase is the dominant factor in determining the total flight range.

After gliding to the specified altitude, Plan C enters the RA-phase, while Plan A and Plan B enter the DI-phase. In the RA-phase, the ramjet engine is started and the vehicle is propelled to fly in dive-accelerate-climb strategy according to the given optimization indicator of maximum terminal velocity. In addition, during the operation of the ramjet, the variation of inlet safety margin is shown in Fig. 9(e). Since the ramjet optimization model used in the assumption is continuously differentiable and the acceleration process pursues the maximum engine thrust, the fuel equivalent ratio is kept at its maximum within a reasonable range. And the safety margin shows a smooth upward trend with the increase of flight Mach number, which is consistent with the characteristic law. In the full RA-phase, the minimum safety margin value is 0.23, which does not cross the safety boundary, and the safe operation of the ramjet is ensured during the optimization process. At the end of this phase, the ramjet propels the vehicle to an altitude of 30 km and increases the flight velocity to 1563.89 m/s, which is a 73.81 % increase compared to the starting velocity. In addition, RA-phase provides the vehicle with an additional 186.52 km flight range increment, reducing the gap with Plan B. After the ramjet fuel is exhausted, the vehicle enters the DI-phase. The flight-path angle is rapidly decreased to reduce the aerodynamic drag in the dive process, while the flight altitude is rapidly decreased to get close to the ground attack target. In order to increase the terminal impact velocity as much as possible, the vehicle drop rapidly in this phase, and the flight ranges are 110.58 km, 50.15 km and 97.51 km respectively, accounting for only 2.07 %, 1.13 % and 2.25 % of the total flight range. Ultimately, Plan C consistently maintains a much higher flight velocity in the DI-phase than the other plans.

Variable geometry RA-phase trajectory optimization

According to the above analysis, it is known that the thrust performance of the ramjet engine directly determines the terminal velocity of the RA-phase and affects the flight velocity and terminal velocity in the DI-phase, and then affects the defense penetration and attack capability of the vehicle. Usually, the closer the operating state of the engine is to the safety boundary, the more its performance can be fully utilized [3]. From the curve of inlet safety margin in RA-phase shown in Fig. 9(e), it can be seen that although the ramjet engine always operates within the safe range, its margin value is large and the minimum value is kept above 0.2. The ramjet operates in an overly conservative state. Although the thrust requirements of the acceleration process keep the fuel equivalent ratio at the maximum, the engine still has a geometrically adjusted performance margin. The engine geometry limits both its operating range and thrust performance. On the one hand, when the Mach number is low, the ramjet needs a large nozzle throat area to ensure the smooth start of the inlet and the safe operation of the engine. On the other hand, when the Mach number gradually increases, the engine needs to reduce the nozzle throat area to build the inlet compression process with high total pressure recovery coefficient and improve the engine thrust performance. In order to give full play to the performance of the ramjet within the safe operating range, this study considers ramjets with different geometric configurations and a variable geometry ramjet as the propulsion plants of the X-43A vehicle respectively, and trajectory optimization for the RA-phase is carried out. Table 6 shows the specific constraint values of the vehicle state in RA-phase. And the geometric parameters of ramjet in each case are given in Table 7. Among them, Case 3 is the geometric configuration considered in the previous study. Therefore, this study takes Case 3 as the reference to analyze the trajectory optimization results in different cases.

Considering the same initial conditions, constraints and optimization indicators, the trajectory optimization of RA-phase is carried out respectively for ramjet cases with different geometric configurations. The changing processes of vehicle state in each case are shown in Fig. 10. In each case, the vehicle first accelerates from a dive at an altitude of 25 km with an initial velocity of 900 m/s, and then starts to climb to the target altitude after diving to an altitude near 18 km. In the dive process, the flight range, altitude and flight-path angle of each case are similar. According to Eq. (1), the altitude change rate of the vehicle depends on the flight velocity and flight-path angle. During the dive, when the flight-path angle drops to the minimum limit value of -10°, it will remain unchanged until the demand for lowering the altitude is slowed down. Then it will turn into an upward trend and rise to 0° at the lowest altitude. After that, the vehicle enters the climbing process, and the differences between each case are gradually obvious. Table 8 shows the terminal state values of the vehicle in different cases. Among them, as the nozzle throat area decreases, the flight velocity of the vehicle at the end of the RA-phase gradually increases. Compared with Case 3, the terminal velocity of Case 1 with a large nozzle throat area decreases by 1.92 %, and the terminal velocity of Case 5 with a small nozzle throat area increases by 2.03 %. As the nozzle throat area decreases, the flight velocity of the vehicle in the RA-phase increases. From Case 1 to Case 5, the nozzle throat area decreases by 0.008 m^2, and the terminal flight velocity increases by 61.73 m/s. Case Ath uses a variable geometry ramjet engine, which have higher acceleration during the climb, and its terminal flight velocity reaches 1675.67 m/s. This is an increase of 142.98 m/s and an improvement of 9.33 % compared to Case 1. In addition, by improving the overall flight velocity, the flight range of Case Ath in the RA-phase is increased by 5.26 km, and the flight time in the RA-phase is reduced by 4.14 s.

Table 6
The specific constraint values of the vehicle state in RA-phase.

State	Initial state	Constraint
Velocity [m/s]	900	[900, 1500]
Altitude [km]	25	[18,35]
Flight-path angle [deg]	0	[−10, 10]
Angle of attack [deg]	2	[−4, 8]

Table 7
Nozzle throat area of ramjet in each case.

Case	Nozzle throat area [m^2]
1	0.039
2	0.037
3	0.035
4	0.033
5	0.031
Ath	Variable

Fig. 11 shows the variation of engine state in the optimization results for different cases. In the RA-phase, the angle of attack, fuel equivalent ratio and nozzle throat area are used as control variables to adjust the operating state of the vehicle. At the initial moment, the thrust output performance of the engine gradually improves with the reduction of the nozzle throat area in each case. During subsequent dive and acceleration, a higher acceleration will lead to a higher velocity increase. When diving to the lowest position, the thrust performance difference of the ramjet reaches the largest. From Case 1 to Case 5, the maximum thrust in RA-phase is increased by 2193.70 N, in which Case 1 is reduced by 4.58 % compared with Case 3, and Case 5 is increased by 4.71 % compared with Case 3. To ensure the normal operation of the ramjet, the minimum value of the margin is limited to 0.05 in the optimization process to avoid the inlet unstart caused by uncertain disturbance. Cases 1–5 all meet the requirements of safe operation, and with the continuous reduction of the nozzle throat area, the inlet safety margin in the initial state is continuously reduced. For a ramjet with fixed geometry, the nozzle throat area should not be less than the corresponding value in Case 5, otherwise, the ramjet cannot start to work from the initial state.

However, under variable geometry condition, the ramjet can automatically adjust the nozzle throat area to ensure that it always operates near the safety boundary and gives full play to its performance until the minimum adjustment boundary of the area is reached. As shown in Fig. 11(d), the nozzle throat area tends to decrease approximately linearly over time within its adjustable range to ensure that the margin is maintained at the minimum limit value. Once the nozzle throat area is adjusted to the minimum value of 0.02 m^2, it will remain at this state, and the margin value will then begin to increase as the velocity increases. The variable geometry adjustment results in a 17.39 % maximum thrust increase in Case Ath compared to Case 3. In addition, in the dive-accelerate-climb strategy, the aerodynamic lift of the vehicle increases rapidly during dive and reaches the maximum at the lowest dive point, as shown in Fig. 12. In Case 5, the maximum aerodynamic lift is increased by 846.60 N (2.96 %) compared with case 3, while in Case Ath, it is increased to 3833.30 N (13.40 %) compared with the reference value. The variable geometry case significantly improves the maximum aerodynamic lift at the end of the dive. Considering that the increase of flight velocity and air density will also lead to the increase of aerodynamic drag, Fig. 12(d) compares the lift-drag ratio in different cases. At the end of the dive, the lift-drag ratio in Case 3 is 2.94, while that in Case Ath is 3.25, an increase of 10.69 %.

Under the influence of aerodynamic lift, thrust, aerodynamic drag and boundary constraints, the adjustment processes of the angle of attack in the diving are shown in Fig. 10(d). Among them, the turn that

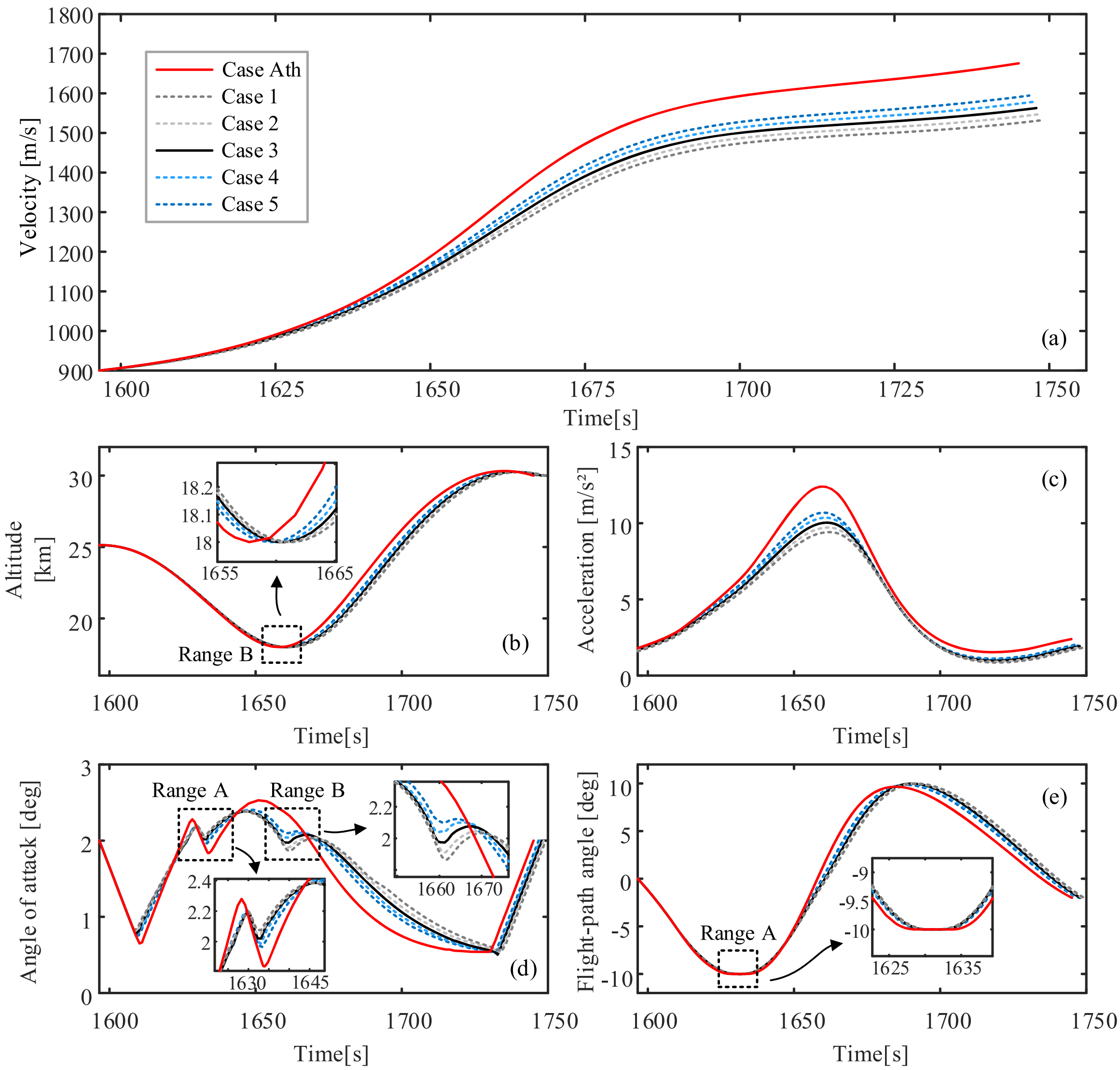


Fig. 10. The changing processes of vehicle state in the optimization results for different cases.

Table 8
The terminal state values of RA-phase in different cases.

case	Velocity [m/s]	Flight Range [km]	Time [s]
1	1532.69	191.85	1749.19
2	1547.54	192.25	1748.54
3	1562.75	192.67	1747.91
4	1578.36	193.20	1747.32
5	1594.42	193.80	1746.77
Ath	1675.67	197.11	1745.05

occurs in Range A is caused by the flight-path angle constraint, while the sudden turn in Range B is caused by the flight altitude constraint. From Eq. (1), when the flight-path angle reaches the constraint and remains unchanged, the angle of attack needs to be appropriately reduced in order to continue to increase the acceleration of the vehicle during the dive. When the vehicle is about to reach the altitude constraint, the acceleration state needs to be slowed down, so this adjustment is realized by increasing the angle of attack. After ending the dive, the flight-path angle of the vehicle shifts to the positive value, and the vehicle enters the climb process. Due to the rapid increase in altitude, the thrust, aerodynamic lift and drag of the vehicle shows a rapid decline trend. The acceleration also tends to decrease, and the velocity increase of the vehicle gradually became relatively gentle. However, on the one hand, the thrust performance of the variable geometry ramjet engine is always better than that of the fixed geometry engine. On the other hand, the advantage of performance makes the vehicle at a higher flight altitude and lower aerodynamic drag at the same moment. Therefore, the acceleration of Case Ath is always in a higher state than other cases.

Conclusion

In this study, an integrated flight/propulsion model that can reflect the engine thrust and safety performance was constructed as the study object, and the trajectory optimization results for the whole flight phase with powered/unpowered ramjet and under different vehicle plans were analyzed, the terminal velocity of hypersonic vehicle constrained by the safety boundary of variable geometry ramjet was discussed. The main results and conclusions of this study are as follows.

(1) This study clarified the superiority of the dive-accelerate-climb strategy adopted by the ramjet engine during its operation process, and verified the significant influence of the constraints on the optimization results. The optimized RA-phase trajectories show significant differences as the lower limit of the altitude constraint is gradually reduced. In the dive process, the maximum

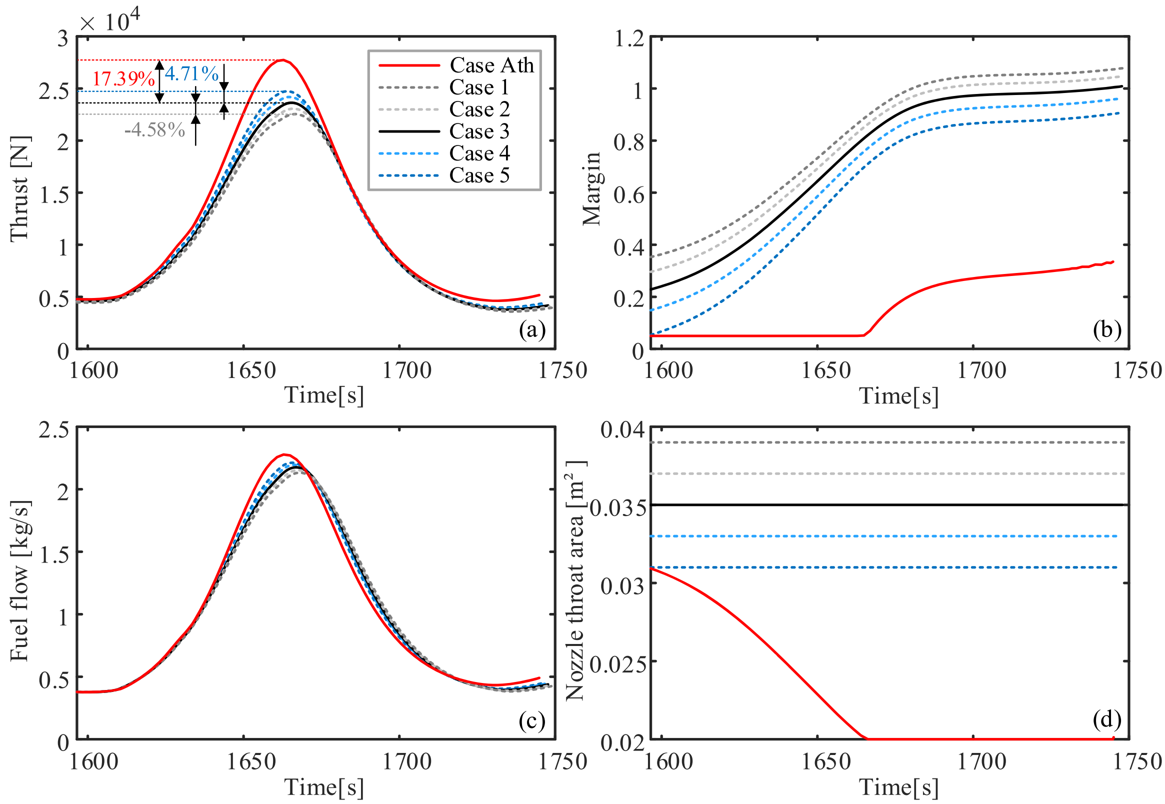


Fig. 11. The variation of engine state in the optimization results for different cases.

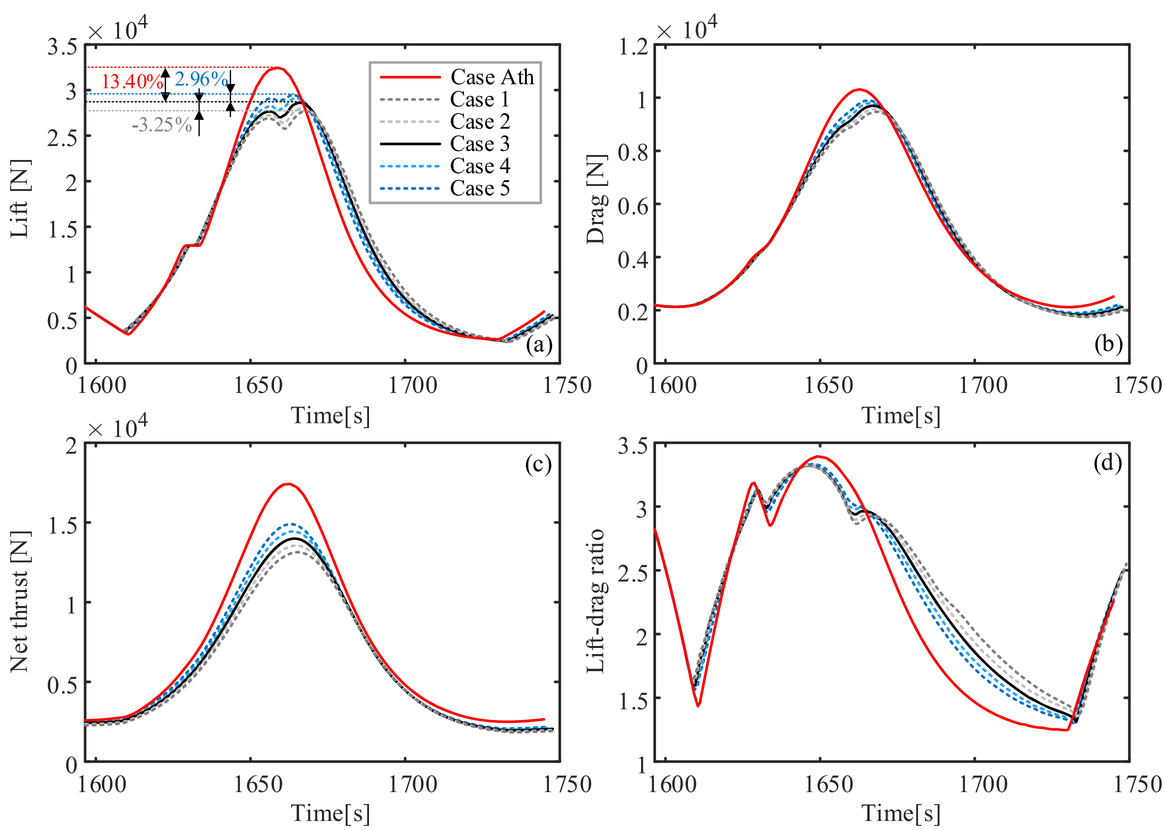


Fig. 12. The variation of the force on the vehicle in the optimization results for different cases.

engine thrust and the maximum aerodynamic lift are increased by 54.05 % and 43.76 %, respectively, compared with the horizontal acceleration process, which significantly improves the acceleration and climb capability of the vehicle in the RA-phase. With the same fuel consumption, the terminal flight velocity of the dive acceleration strategy is increased by 0.74 % compared with the horizontal acceleration strategy, and the flight time is reduced by 21.16 s. The dive-acceleration-climb strategy helps to improve the velocity of the vehicle.

(2) Plan C, powered by the X43-A, can increase the terminal flight velocity of the hypersonic vehicle. In each optimization result, GL-phase adopts the skip-glide trajectory, which can account for more than 80 % of the total flight range, and is the dominant factor in determining the total flight range. In unpowered plans, the flight range of CAV-H is 908.83 km longer than that of X-43A by its lighter mass and high lift-drag ratio aerodynamic shape, but it has no significant effect on the terminal impact velocity. In the powered Plan C, the RA-phase increases the terminal velocity of the X-43A vehicle by 159.51 % by increasing the initial altitude and velocity of the DI-phase. The employment of a powered RA-phase can significantly improve the vehicle's defense penetration capability.

(3) The variable geometry gives full play to the performance of the ramjet within the safety boundary and further improves the terminal velocity of the powered flight plan. In the trajectory optimization results for RA-phase with different geometric parameters of ramjet, it is difficult for the ramjet to balance the thrust performance and safety performance in the large envelope operating range due to a fixed geometry. In the process of diving, the variable geometry ramjet keeps the inlet margin near the safety boundary by adjusting the nozzle throat area. Compared with the reference geometry condition, the maximum thrust of the variable geometry ramjet is increased by 17.39 %, the acceleration and aerodynamic lift of the vehicle are improved, and the terminal velocity of the RA-phase is increased by 9.33 %.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgment

This research work is supported by the National Natural Science Foundation of China (Grant No. 52125603).

Appendix A

In order to take the influence of nozzle throat area into consideration in the ramjet optimization model, the coefficients in Eq. (10) and (11) are obtained by fitting the 4th order polynomial of nozzle throat area, as shown in Eq. (A1). Table A1 and Table A2 provide the fitting results for each coefficient in Eq. (10), (11).

$$k_{ij} = \sum_{n=1}^{5} p_n A_{th}^{5-n} \tag{A1}$$

Table A1
Fitting results for thrust fitting coefficients.

k_{Fij}	p_1	p_2	p_3	p_4	p_5
k_{F00}	−8.06e8	1.006e8	−4.564e6	8.729e4	−582.5
k_{F10}	1.866e8	−2.343e7	1.071e6	−2.071e4	137.6
k_{F01}	−6.956e9	8.598e8	−3.753e7	5.416e5	1.65e4
k_{F11}	1.087e9	−1.369e8	6.272e5	−1.159e5	−638.4
k_{F02}	2.628e9	−3.362e8	1.569e7	−3.108e5	2375

Table A2
Fitting results for margin fitting coefficients.

$k_{\xi ij}$	p_1	p_2	p_3	p_4	p_5
$k_{\xi 00}$	−3.543e6	3.857e5	−1.86e4	533	−7.283
$k_{\xi 10}$	−3.44e6	4.868e5	−2.419e4	474.8	−1.491
$k_{\xi 01}$	2.751e7	−3.118e6	1.295e5	−2219	7.572
$k_{\xi 20}$	7.471e5	−1.248e5	7053	−157.1	0.7891
$k_{\xi 11}$	4.711e5	−5.289e5	2.245e4	−460.7	5.292
$k_{\xi 02}$	−6.719e7	7.71e6	−3.299e5	6187	−40.99
$k_{\xi 30}$	2.553e4	2840	−385	11.6	−0.05645
$k_{\xi 21}$	−2.493e6	2.94e5	−1.306e4	260.3	−2.128
$k_{\xi 12}$	9.966e6	−1.225e6	5.656e4	−1143	8.162
$k_{\xi 03}$	3.402e7	−3.736e6	1.523e5	−2712	17.34
$k_{\xi 40}$	9534	−1219	58.96	−1.229	0.006955

(continued on next page)

Table A2 (*continued*)

$k_{\xi ij}$	p_1	p_2	p_3	p_4	p_5
$k_{\xi 31}$	−1.371e5	1.179e4	−329.1	2.676	0.01577
$k_{\xi 22}$	1.889e6	−1.913e5	7072	−112.1	0.6461
$k_{\xi 13}$	−8.883e6	9.702e5	−3.947e4	703.8	−4.559

References

[1] S. Luo, Y. Sun, J. Liu, Performance analysis of the hypersonic vehicle with dorsal and ventral intake, Aerosp. Sci. Technol. 131 (2022), 107964, https://doi.org/10.1016/j.ast.2022.107964.
[2] Y. Zhu, W. Peng, R. Xu, P. Jiang, Review on active thermal protection and its heat transfer for airbreathing hypersonic vehicles, Chin. J. Aeronaut. 31 (10) (2018) 1929–1953, https://doi.org/10.1016/j.cja.2018.06.011.
[3] C. Lv, J. Chang, W. Bao, D. Yu, Recent research progress on airbreathing aero-engine control algorithm, Propulsion Power Res. 11 (1) (2022) 1–57, https://doi.org/10.1016/j.jppr.2022.02.003.
[4] C. McClinton, V. Rausch, L. Nguyen, J. Sitz, Preliminary X-43 flight test results, Acta Astronaut. 57 (2005) 266–276, https://doi.org/10.1016/j.actaastro.2005.03.060.
[5] Y. Liu, D. Mo, Y. Wu, Design of hypersonic variable cycle turboramjet engine based on hydrogen fuel aiming at 2060 carbon neutralization, in: Proceedings of the 2021 Asia-Pacific International Symposium on Aerospace Technology, 2021, pp. 1183–1197, https://doi.org/10.1007/978-981-19-2635-8_85.
[6] D. Han, J. Kim, K. Kim, Conjugate thermal analysis of X-51A-like aircraft with regenerative cooling channels, Aerosp. Sci. Technol. 126 (2022), 107614, https://doi.org/10.1016/j.ast.2022.107614.
[7] M. Abrams, The hypersonic missile gap, Mech. Eng. 145 (2) (2023) 40–45, https://doi.org/10.1115/1.2023-Mar2.
[8] Y. Ding, X. Yue, G. Chen, J. Si, Review of control and guidance technology on hypersonic vehicle, Chin. J. Aeronaut. 35 (7) (2022) 1–18, https://doi.org/10.1016/j.cja.2021.10.037.
[9] D. Sziroczak, H. Smith, A review of design issues specific to hypersonic flight vehicles, Prog. Aerosp. Sci. 84 (2016) 1–28, https://doi.org/10.1016/j.paerosci.2016.04.001.
[10] T. Zhang, X. Yan, W. Huang, X. Che, Z. Wang, E. Lu, Design and analysis of the air-breathing aircraft with the full-body wave-ride performance, Aerosp. Sci. Technol. 119 (2021), 107133, https://doi.org/10.1016/j.ast.2021.107133.
[11] S. Zhai, J. Yang, Piecewise analytic optimized ascent trajectory design and robust adaptive finite-time tracking control for hypersonic boost-glide vehicle, J. Franklin Inst. 357 (9) (2020) 5485–5501, https://doi.org/10.1016/j.jfranklin.2020.03.002.
[12] M. Dileep, S. Kamath, V. Nair, Particle swarm optimization applied to ascent phase launch vehicle trajectory optimization problem, Procedia Comput. Sci. 54 (2015) 516–522, https://doi.org/10.1016/j.procs.2015.06.059.
[13] Z. Li, T. Yang, Z. Feng, The multi-objective trajectory optimization for hypersonic glide vehicle based on normal boundary intersection method, Math. Probl. Eng. 9407238 (2016), https://doi.org/10.1155/2016/9407238.
[14] Z. Shen, J. Yu, X. Dong, Y. Hua, Z. Ren, Penetration trajectory optimization for the hypersonic gliding vehicle encountering two interceptors, Aerosp. Sci. Technol. 121 (2022), 107363, https://doi.org/10.1016/j.ast.2022.107363.
[15] J. He, Y. Liu, S. Li, Y. Tang, Minimum-fuel ascent of hypersonic vehicle considering control constraint using the improved pigeon-inspired optimization algorithm, Int. J. Aerosp. Eng. (2020), 3024607, https://doi.org/10.1155/2020/3024607.
[16] B. Yan, R. Liu, P. Dai, M. Xing, S. Liu, A rapid penetration trajectory optimization method for hypersonic vehicles, Int. J. Aerosp. Eng. (2019), 1490342, https://doi.org/10.1155/2019/1490342.
[17] S. Yang, T. Cui, X. Hao, D. Yu, Trajectory optimization for a ramjet-powered vehicle in ascent phase via the Gauss pseudo-spectral method, Aerosp. Sci. Technol. 67 (2017) 88–95, https://doi.org/10.1016/j.ast.2017.04.001.
[18] C. Chuang, H. Morimoto, Periodic optimal cruise for a hypersonic vehicle with constraints, J. Spacecr. Rockets 34 (2) (1997) 165–171, https://doi.org/10.2514/2.3205.
[19] J. Song, H. Choi, Hybrid Control Trajectory Optimization for Air-breathing Hypersonic Vehicle, IFAC-PapersOnLine 53 (2) (2020) 14768–14774, https://doi.org/10.1016/j.ifacol.2020.12.1900.
[20] J. Zhao, R. Zhou, Reentry trajectory optimization for hypersonic vehicle satisfying complex constraints, Chin. J. Aeronaut. 26 (6) (2013) 1544–1553, https://doi.org/10.1016/j.cja.2013.10.009.
[21] J. Zheng, J. Chang, S. Yang, X. Hao, D. Yu, Trajectory optimization for a TBCC-powered supersonic vehicle with transition thrust pinch, Aerosp. Sci. Technol. 84 (2019) 214–222, https://doi.org/10.1016/j.ast.2018.10.026.
[22] G. Huang, Y. Lu, Y. Nan, A survey of numerical algorithms for trajectory optimization of flight vehicles, Sci. China: Technol. Sci. 55 (9) (2012) 2538–2560, https://doi.org/10.1007/s11431-012-4946-y.
[23] Y. Wu, J. Deng, L. Li, X. Su, L. Li, A hybrid particle swarm optimization-gauss pseudo method for reentry trajectory optimization of hypersonic vehicle with navigation information model, Aerosp. Sci. Technol. 118 (2021), 107046, https://doi.org/10.1016/j.ast.2021.107046.
[24] J. Betts, Survey of numerical methods for trajectory optimization, J. Guid. Control Dyn. 21 (2) (1998) 193–207, https://doi.org/10.2514/2.4231.
[25] C. Guo, J. Zhang, Y. Luo, L. Yang, Phase-matching homotopic method for indirect optimization of long-duration low-thrust trajectories, Adv. Space Res. 62 (3) (2018) 568–579, https://doi.org/10.1016/j.asr.2018.05.007.
[26] R. Chai, A. Savvaris, A. Tsourdos, S. Chai, Y. Xia, A review of optimization techniques in spacecraft flight trajectory design, Prog. Aerosp. Sci. 109 (2019), 100543, https://doi.org/10.1016/j.paerosci.2019.05.003.
[27] K. An, Z. Guo, X. Xu, W. Huang, A framework of trajectory design and optimization for the hypersonic gliding vehicle, Aerosp. Sci. Technol. 106 (2020), 106110, https://doi.org/10.1016/j.ast.2020.106110.
[28] P. Dai, D. Feng, W. Feng, J. Cui, L. Zhang, Entry trajectory optimization for hypersonic vehicles based on convex programming and neural network, Aerosp. Sci. Technol. 137 (2023), 108259, https://doi.org/10.1016/j.ast.2023.108259.
[29] X. Liu, Research On Trajectory Design and Guidance Technology of Boost-Glide Vehicle, National University of Defense Technology, Changsha China, 2012.
[30] T. Cui, S. Tang, C. Zhang, D. Yu, Hysteresis phenomenon of mode transition in ramjet engines and its topological rules, J. Propul. Power 28 (6) (2012) 1277–1284, https://doi.org/10.2514/1.B34419.
[31] C. Lv, Q. Huang, J. Chang, Z. Wang, J. Zheng, D. Yu, Mode transition path optimization for turbine-based combined-cycle ramjet stage under uncertainty propagation of integrated airframe-propulsion system, Energy 268 (2023), 126718, https://doi.org/10.1016/j.energy.2023.126718.
[32] S. Wei, Y. Zou, F. Sun, O. Christopher, A pseudo-spectral method for solving optimal control problem of a hybrid tracked vehicle, Appl. Energy 194 (15) (2017) 588–595, https://doi.org/10.1016/j.apenergy.2016.07.020.
[33] I. Ross, M. Karpenko, A review of pseudo-spectral optimal control: from theory to flight, Annu. Rev. Control 36 (2) (2012) 182–197, https://doi.org/10.1016/j.arcontrol.2012.09.002.
[34] G. Liu, B. Li, Y. Ji, A modified HP-adaptive pseudo-spectral method for multi-UAV formation reconfiguration, ISA Trans. 129 (2022) 217–229, https://doi.org/10.1016/j.isatra.2022.01.015.
[35] A. Yazdani, K. Sammut, O. Yakimenko, A. Lammas, Feasibility analysis of using the hp-adaptive Radau pseudo-spectral method for minimum-effort collision-free docking operations of AUV, Rob Auton. Syst. 133 (2020), 103641, https://doi.org/10.1016/j.robot.2020.103641.
[36] P. Lu, B. Pan, Highly constrained optimal launch ascent guidance, J. Guid. Control Dyn. 33 (2) (2010) 404–414, https://doi.org/10.2514/1.45632.